

Eric I Ramirez

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EDUCATION

California State University, Chico

Expected graduation: May 2026

Bachelor of Science in **Computer Engineering**

Bilingual (English/Spanish)

SKILLS

- **Developing** – SystemVerilog, VHDL, Vivado, Quartus Prime, FPGA Development, FreeRTOS, ROS2, OpenCV
- **Hardware** – Embedded Hardware Integration, Circuit Prototyping, Soldering, Logic Analyzer, Oscilloscope
- **Software** – Embedded C, C++, Assembly, Python, LTSpice, Autodesk(EAGLE), Uvision(KEIL), MATLAB, Git

RESEARCH & DEVELOPMENT

Undergraduate Researcher, College of Engineering (CSU Chico)

May 2025 - Aug 2025

- Developed a person-following robot using ROS2, Python/OpenCV, achieving real-time LED tracking at 30 FPS.
- Integrated Raspberry Pi 4/OpenCR for synchronized vision-motor control, enabling stable tracking at 0.2-0.4 m/s.
- Implemented HSV masking, morphological filtering, and a ± 15 px safe zone to reduce jitter and overshoot.
- Published real-time overlays and masks in ROS2 topics, cutting tuning and debug time by 40%.

EMBEDDED SYSTEMS PROJECTS

Real-Time Temperature Monitoring System, College of Engineering (CSU Chico)

Apr 2025 – May 2025

- Programmed Tiva Microcontroller to sample analog input with a 12-bit ADC and convert sensor data to $^{\circ}\text{C}/^{\circ}\text{F}$.
- Configured SysTick interrupts at 16MHz for periodic sampling and LCD updates, ensuring real-time display.
- Implemented fixed-point arithmetic for efficient computation, achieving $\pm 0.1^{\circ}\text{C}$ resolution without floating-point.
- Used UART at 57,600 bps to send serial raw temp data between two Tiva boards, later formatted for PC display.

Control System for DC Motor, College of Engineering (CSU Chico)

Feb 2025 - Mar 2025

- Programmed SysTick PWM with interrupts to run a DC motor at 20%, 50%, and 80% speeds for precise control.
- Optimized microcontroller to 40 MHz and rewrote ISR, enabling real-time updates verified by logic analyzer tests.
- Eliminated manual delay handling, enhancing speed control accuracy and reducing overall testing time by 30%.

TECHNICAL PROJECTS

Three-Stage BJT Audio Amplifier, College of Engineering (CSU Chico)

Mar 2025 - May 2025

- Designed and simulated a cascaded BJT amplifier in LTSpice with stage gains of -15.6, -16.8, and unity for buffer.
- Built and tested breadboard circuits with teammate, achieving overall gain of 118x (41 dB) into an 8 ohm speaker.
- Applied biasing, degeneration, and tuning to fix clipping and noise, delivering clear audio from 20Hz–20KHz.

Arduino Shield PCB Design, College of Engineering (CSU Chico)

Sep 2024 – Dec 2024

- Designed a unified Arduino shield schematic in Autodesk (EAGLE) by combining multiple circuit sections together.
- Created a PCB layout within the shield template, placing components per the BOM and meeting specifications.
- Integrated thermocouple, BCD converter, buzzer, pushbutton into one design for an Arduino shield.

DC Motor Design, College of Engineering (CSU Chico)

Nov 2023 – Dec 2023

- Engineered a DC motor by upscaling a factory model 350% using a 3D-printed shaft, commutator, and housing.
- Redesigned coil winding based on electric field alignment to ensure sequential polarity with the magnets.
- Achieved continuous 3000 RPM through enhancing electrical contact to sustain efficient motor performance.

ADDITIONAL EXPERIENCE

Computer Assembly Technician, Entrepreneur - Redding, CA

Apr 2019 – Present

- Built 10+ custom PC builds tailored to user budgets ranging from \$400 to \$2600.
- Troubleshoot BIOS, power, and hardware integration issues for optimal performance and efficiency.

Electronics Repair Technician, Entrepreneur - Redding, CA

Jan 2017 - Present

- Repaired computers, laptops, tablets, and smartphones, including screen, battery, and camera module repairs.
- Diagnosed and resolved hardware and software issues across multiple devices to ensure reliable performance.